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Experiments on Fraud Detection use case with QML and TDA Mapper

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***Abstract*— In the era of online financial transactions, it is significant for the credit card firms to be equipped with capabilities to identify fraudulent credit card transactions. This work covers study and implementation of two approaches for developing a credit card fraud detection model. First one, with hybrid quantum neural networks. In recent times, Quantum Computers (QC) are making their footprints into AI/ML domain. Quantum neural networks (QNN) hybrid with classical neural net has been used in various tasks such as – natural language processing, image processing etc. The second approach is with Topological Data Analysis (TDA). Finding topological structure in the input data also become relevant from the perspective of noise reduction. The visualization capabilities of TDA can become an aid in classification of credit card fraud as well. TDA is implemented with mapper based method here. In hybrid QNN, we are covering a reference implementation of Xanadu's StrawberryFields, where a classical network processes the input to be fed into a QNN model. Although technique wise these two approaches are drastically different, for the sake of generalization we implement TDA and hybrid QNN with a publicly available credit card fraud detection dataset. We tested with balanced fraud and genuine features and hybrid QNN model provides accuracy of 89.5%, whereas TDA mapper with our novel approach of classification provides an accuracy of 94%.**

Keywords—QML, QNN, TDA Mapper, StrawberryFields, Fraud Detection, Classification

I. POSTER RELEVANCE

This poster presents the two approaches, hybrid QNN and TDA for credit card fraud detection. It compares both of these approaches with a publicly available fraud detection dataset. We provide a comparative study between hybrid QNN and TDA.

II. POSTER : EXPERIMENT ON FRAUD DETECTION USE CASE WITH QML AND TDA

In current economy, various credit card companies want to track fraud transaction to provide better reliability. The basic motivation behind these study is to implement an accurate and precise approach for detecting credit card frauds. We present two approaches, one with hybrid QNN and other one is with TDA. The implemented hybrid QML model, which incorporated in association of both classical and quantum neural network is covering a reference implementation of [1]. Here, initially the input data pass through a classical neural network which transform 10 input to 14 output data points which are then used in the first layer of a quantum neural network as gate parameter. With this approach the hurdle of encoding of the input data has been handled. The initial layer of quantum neural network is followed by four standard quantum neural network layers. The implementation is done through Xanadu's Strawberryfields [2].

TDA basically explores the topological structures inside the data, which includes dimensionality reduction, clustering, persistent homology etc.[3], [4]. Here, we implemented TDA with mapper method and find clusters of fraud and genuine transactions. As the classification is binary in nature, we use two clusters in the TDA mapper method. Classification has been done using a novel probabilistic approach, which labels the unknown data points.

III. DATASET

The dataset covers credit card transactions done by European cardholders in September 2013. The dataset contains 492 frauds out of 284,807 transactions that occurred over the course of two days. The dataset is skewed, with the positive class (frauds) accounting for only 0.172 percent of all transactions. The data is made up of 28 features that were created using a PCA transformation [5].

IV. EXPERIMENTATION

Total of 982 data points out of which 492 data are fraud and 492 are genuine. Train Test split is done on 3:1 having 197 unknown points and rest 785 as known data points. For TDA approach we considered, 20 hypercube with 30% overlapping and two clusters has been selected. The same input data is fed into the hybrid QNN model for classification. We used accuracy, precision and recall and ROC as metric to measure performance. The hybrid QNN model shown the accuracy of 89.5% whereas the TDA approach provides accuracy of 94%. The detail result is shown in Table1 below.

Table 1Results of the Experimentations (%)

Approach	Accuracy	Precision	Recall	ROC
Hybrid QNN	0.895	0.90	0.89	0.87
TDA	0.94	0.99	0.98	0.935

Here, we present two approaches for fraud detection with reasonably comparable results. However, these scores came from initial experimentation, with larger and complicated data, combination of both these approaches can generate more appropriate results and improvements over other existing models. In the following poster we presented the architecture, experiments of the approaches.

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